

① a)

$$180^\circ \text{ frequency} = 7 \text{ rad/s} = \omega_{180^\circ}$$

$$\text{Gain at } 7 \text{ rad/s} = -18$$

$$\Rightarrow \text{Gain Margin} = 10^{18/20} = 7.94$$

$$0 \text{ dB frequency} = 2.3 \text{ rad/s} = \omega_{0\text{dB}}$$

$$\text{Phase at } \omega_{0\text{dB}} = -135^\circ \text{ (useful later on)}$$

$$\Rightarrow \text{Phase Margin} = 180^\circ - 135^\circ = 45^\circ$$

$$\text{closed loop bandwidth} = \omega_{-3\text{dB}} = 3 \text{ rad/s}$$

$$b) \text{ Phase Margin} = \tan^{-1} \left(\frac{2\beta}{\sqrt{4\beta^4 + 1} - 2\beta^2} \right) = 45^\circ$$

taking tan on both sides $\Rightarrow$

$$1 = \frac{2\beta}{\sqrt{4\beta^4 + 1} - 2\beta^2}$$

$$\Rightarrow 4\beta^2 = \sqrt{4\beta^4 + 1} - 2\beta^2$$

$$\Rightarrow 6\beta^2 = \sqrt{4\beta^4 + 1}$$

$$\Rightarrow 36\beta^4 = 4\beta^4 + 1$$

$$\Rightarrow \beta^4 = 1/32$$

$$\Rightarrow \beta = 0.42$$

$$\% \text{ OS} = e^{-\beta\pi/\sqrt{1-\beta^2}} \times 100 \% = 23.365 \%$$

$$\text{Band width} = \omega_n \sqrt{1 - 2\zeta^2} + \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

$$3 = \omega_n \sqrt{1 - 2(0.42)^2} + \sqrt{4(0.42)^4 - 4(0.42)^2 + 2}$$

$$3 = \omega_n \sqrt{1 - 0.3528} + \sqrt{0.12446 - 0.7056 + 2}$$

$$3 = \omega_n \sqrt{0.6472} + \sqrt{1.41886}$$

$$3 = \omega_n (1.355861018) \Rightarrow \boxed{\omega_n = 2.2126 \text{ rad/s}}$$

$$T_s = \frac{4}{f\omega_n} = \frac{4}{(0.42)(2.2126)} = \boxed{4.304 \text{ s} = T_s}$$

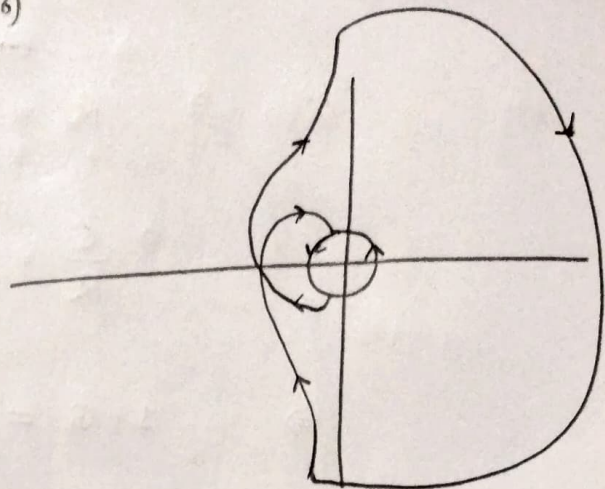
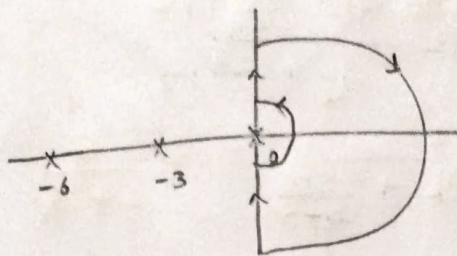
$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{2.2126 \sqrt{1 - (0.42)^2}} = 1.5645 \text{ s}$$

$$\Rightarrow \boxed{T_p = 1.5645 \text{ s}}$$

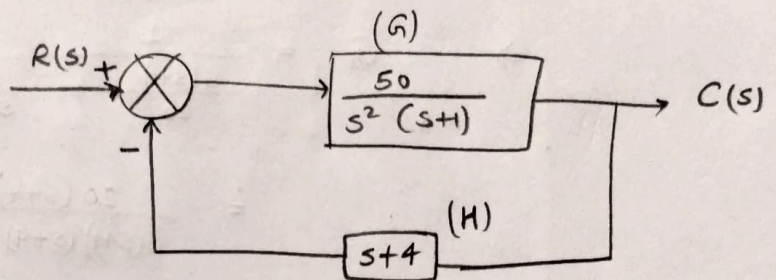
② a.)

$$G(s) = \frac{50}{s(s+3)(s+6)}$$

System 1



b) System 2



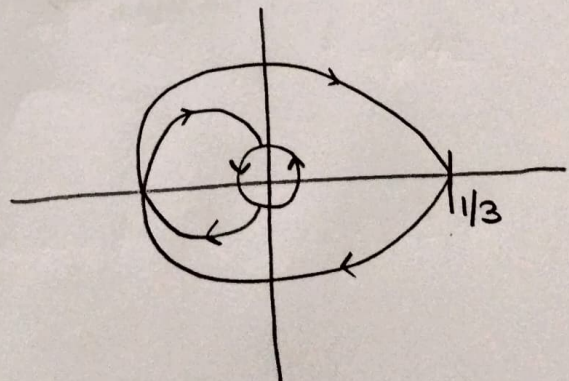
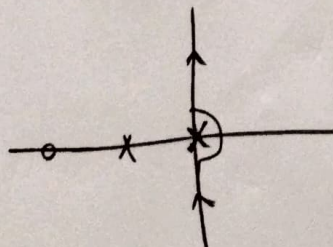
$$(R - HC)G = C$$

$$\Rightarrow RG - HCG = C$$

$$\Rightarrow RG = C(1 + HG)$$

$$\Rightarrow \frac{C}{R} = \frac{G}{1 + HG} = \frac{G}{1 + G'}$$

$$G' = GH = \frac{50(s+4)}{s^2(s+1)}$$



(c) System 3

Like last one,

$$(R - CK)G = C$$

$$\Rightarrow R = \left(\frac{s(s+1)}{20} + \frac{s+3}{s+4} \right) C$$

$$\Rightarrow \frac{C}{R} = \frac{1}{\frac{s(s+1)}{20} + \frac{s+3}{s+4}} = \frac{G}{1+G}$$

$$\Rightarrow 1+G = \left(\frac{s(s+1)}{20} + \frac{s+3}{s+4} \right) G$$

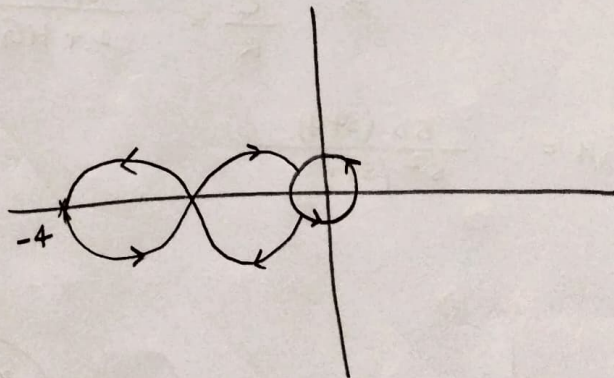
$$\Rightarrow G = \frac{1}{\frac{s(s+1)}{20} - \frac{1}{s+4}}$$

$$= \frac{20(s+4)}{s(s+1)(s+4) - 20}$$

$$= \frac{20(s+4)}{s^3 + 5s^2 + 4s - 20}$$

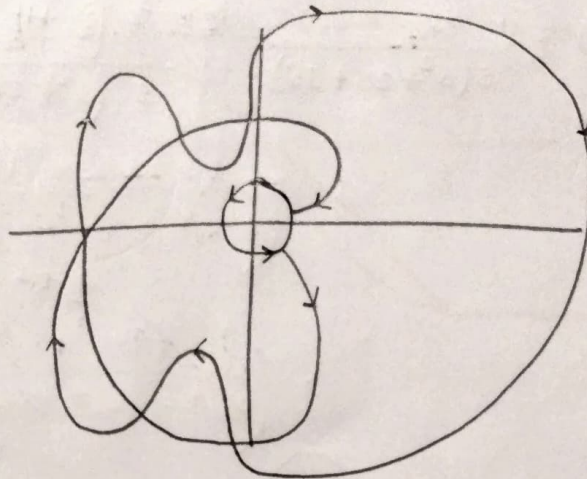
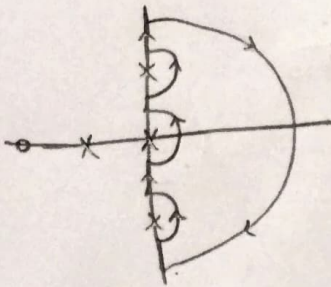
← roots

$$(s-1.4) \left(s - (-3.24 + 1.75j) \right) \left(s - (-3.24 - 1.75j) \right)$$



b) System 4

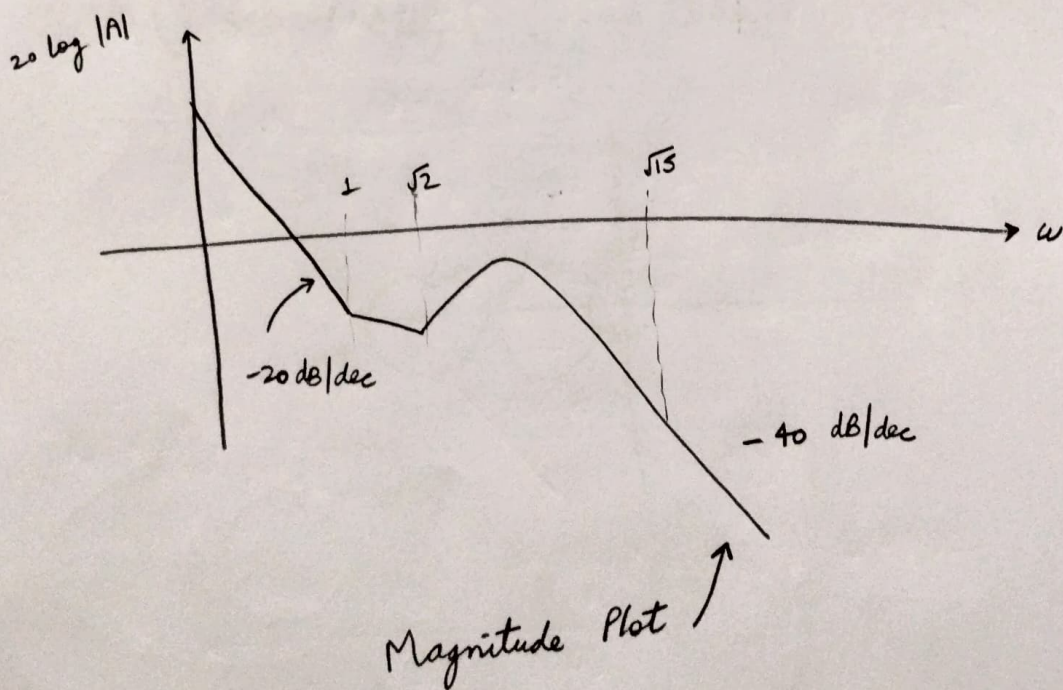
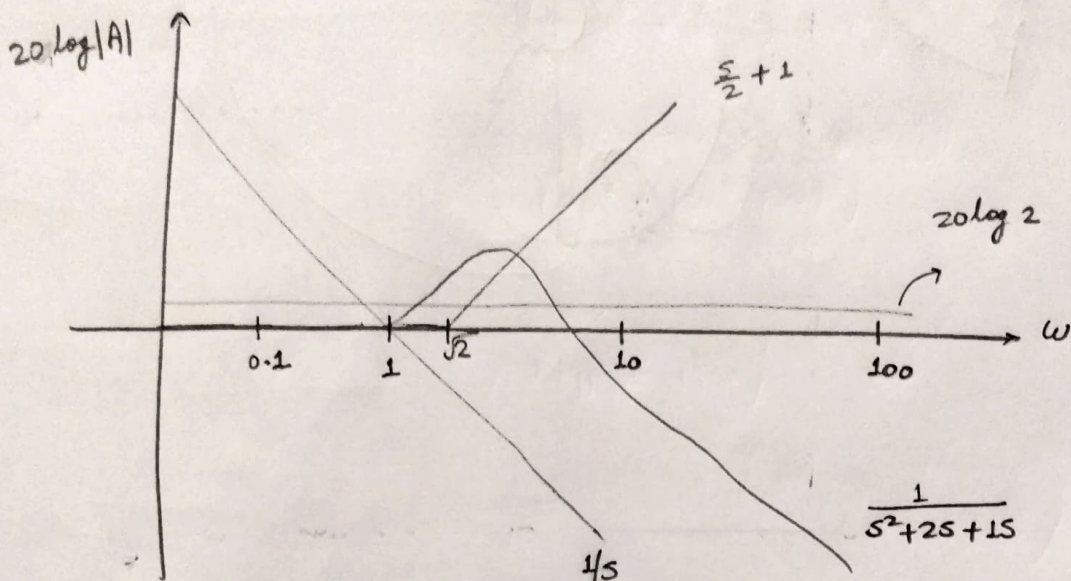
$$G(s) = \frac{100(s+5)}{s(s+3)(s^2+4)}$$

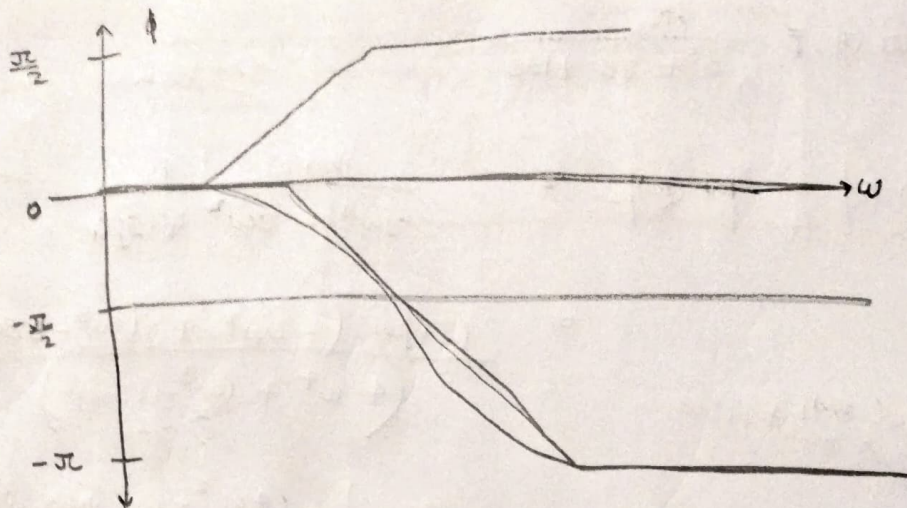


3.

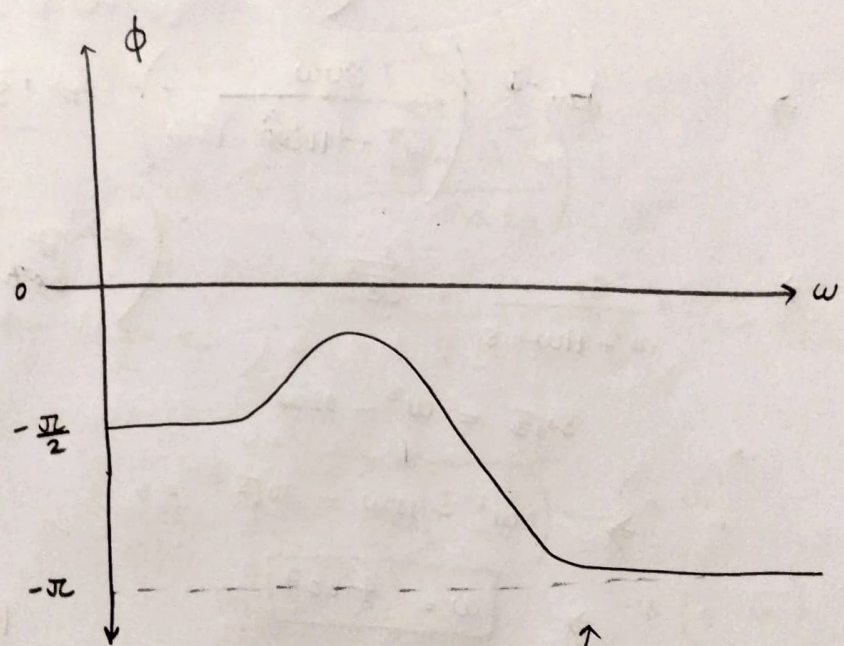
$$L(s) = G_c(s) G(s) = \frac{K(s+2)}{s^3 + 2s^2 + 15s}$$

(a) $G(s) = \frac{s+2}{s(s^2+2s+15)} = \frac{2\left(\frac{s}{2}+1\right)}{s(s^2+2s+15)}$





$\Downarrow$



Phase Plot $\nearrow$

$$(b) \quad G(s) = \frac{s+2}{s^3 + 2s^2 + 15s}$$

$$|G(s)| = |G(j\omega)| = \frac{j\omega + 2}{-j\omega^3 - 2\omega^2 + 15j\omega} \quad \text{--- (1)}$$

$$= \frac{(2+j\omega)(-2\omega^2 + j(\omega^3 - 15\omega))}{(4\omega^4 + (\omega^3 - 15\omega)^2)}$$

$$= \frac{-(30\omega j) - (4\omega^2 + \omega^4 - 15\omega^2)}{(4\omega^4 + (\omega^3 - 15\omega)^2)}$$

$$-4\omega^2 + 2j\omega^3 - 30\omega j - \omega^4 + 15\omega$$

$$\Rightarrow \tan^{-1} \left(\frac{30\omega}{\omega^4 + 11\omega^2 - 15\omega} \right) = 30^\circ$$

$$\Rightarrow \frac{30}{\omega^3 + 11\omega - 15} = \frac{1}{\sqrt{3}} \quad \left(\text{taking tan on both sides} \right)$$

$$\Rightarrow 30\sqrt{3} = \omega^3 - 11\omega$$

$$\Rightarrow \omega^3 - 11\omega - 30\sqrt{3} = 0$$

$$\Rightarrow \boxed{\omega = 4.697}$$

$$|A| = 1$$

$$\omega \text{ in (1)} \Rightarrow K \times 0.092 = 1$$

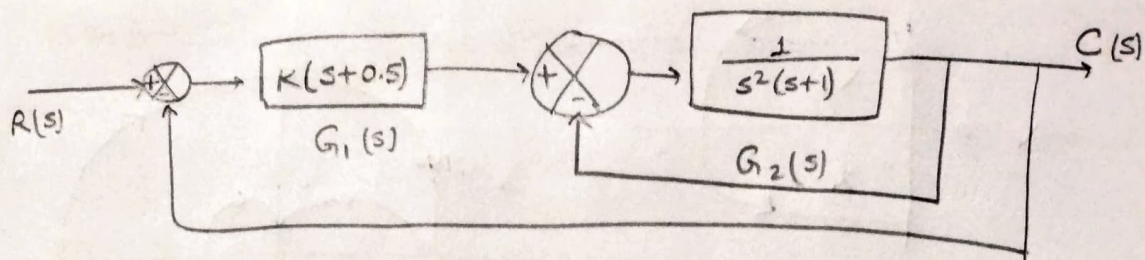
$$\Rightarrow \boxed{K = \frac{1}{0.092} = 10.869}$$

$$K_v = \lim_{s \rightarrow 0} s G(s) = \frac{(10.869)(s+2)}{s(s^2 + 2s + 15)}$$

$$= \frac{10.869 \times 2}{15} = 1.4492$$

$$\boxed{e_{ss} = \frac{1}{K_v} = 0.69}$$

6.



$$((R-C)G_1 - C)G_2 = C$$

$$(R-C)G_1 \frac{C}{R} = \frac{C}{G_2} + C$$

$$RG_1 - CG_1 = C \left(1 + \frac{1}{G_2} \right)$$

$$RG_1 - CG_1 = C \left(\frac{G_2 + 1}{G_2} \right)$$

$$RG_1 = C \left(\frac{G_2 + 1 + G_1 G_2}{G_2} \right)$$

$$\Rightarrow \frac{C}{R} = \frac{G_1 G_2}{G_2 + 1 + G_1 G_2} = \frac{1}{1 + \frac{1}{G_1 G_2} + \frac{1}{G_1}}$$

$$= \frac{1}{1 + \frac{1}{G_1} \left(1 + \frac{1}{G_2} \right)} \rightarrow G'$$

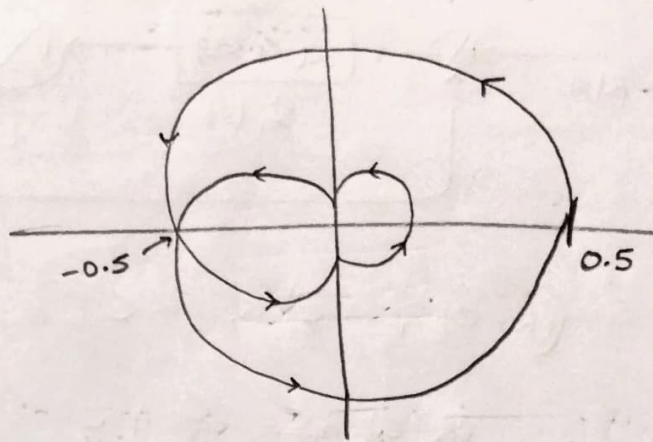
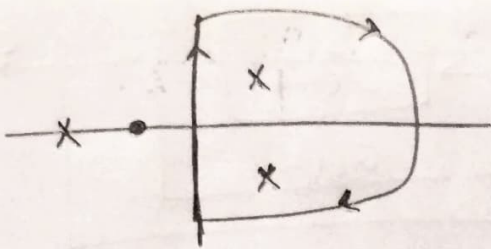
$$= \frac{G_1 \left(\frac{G_2}{1 + G_2} \right)}{1 + G_1 \left(\frac{G_2}{1 + G_2} \right)} \quad (\div G')$$

$$= \frac{G''}{1 + G''} \quad \text{this plot}$$

$$G'' = G_1 \left(\frac{G_2}{1 + G_2} \right) = K(s+0.5) \left(\frac{\frac{1}{s^2(s+1)}}{1 + \frac{1}{s^2(s+1)}} \right)$$

$$= K(s+0.5) \left(\frac{1}{s^2(s+1) + 1} \right)$$

$$= \frac{K(s+0.5)}{(s+1.47)(s-(0.23+0.79j))(s-(0.23-0.79j))}$$



$$s = j\omega \Rightarrow$$

$$G''(j\omega) = \frac{(j\omega + 0.5)}{-\omega^2(j\omega + 1) + 1}$$

$$= \frac{j\omega + 0.5}{(1 - \omega^2) - j\omega^3} = C$$

$$\Rightarrow j\omega + 0.5 = C(1 - \omega^2) - j\omega^3 C$$

$$\Rightarrow j(\omega + \omega^3 C) + (0.5 - C(1 - \omega^2)) = 0$$

$$\omega + \omega^3 C = 0$$

$$0.5 - C(1 - \omega^2) = 0$$

$$\Rightarrow C = \frac{-\omega}{\omega^3}$$

$$C = \frac{0.5}{1 - \omega^2}$$

$$\Rightarrow \frac{-\omega}{\omega^3} = \frac{0.5}{1 - \omega^2}$$

$$\Rightarrow -\omega + \omega^3 = 0.5\omega^3$$

$$\Rightarrow 0.5\omega^3 = \omega \Rightarrow \omega = \pm\sqrt{2}$$

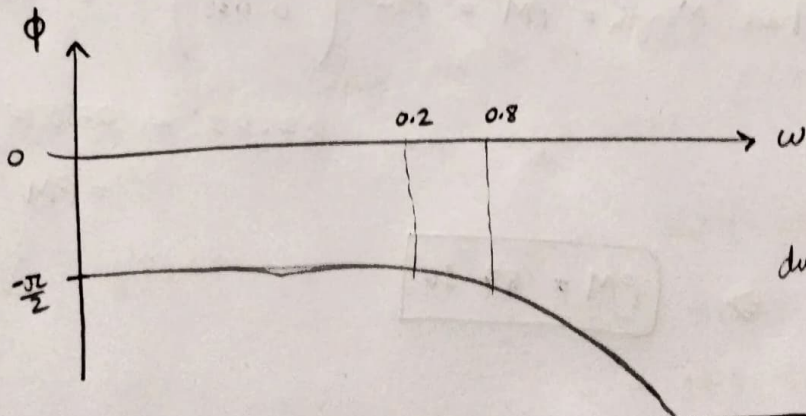
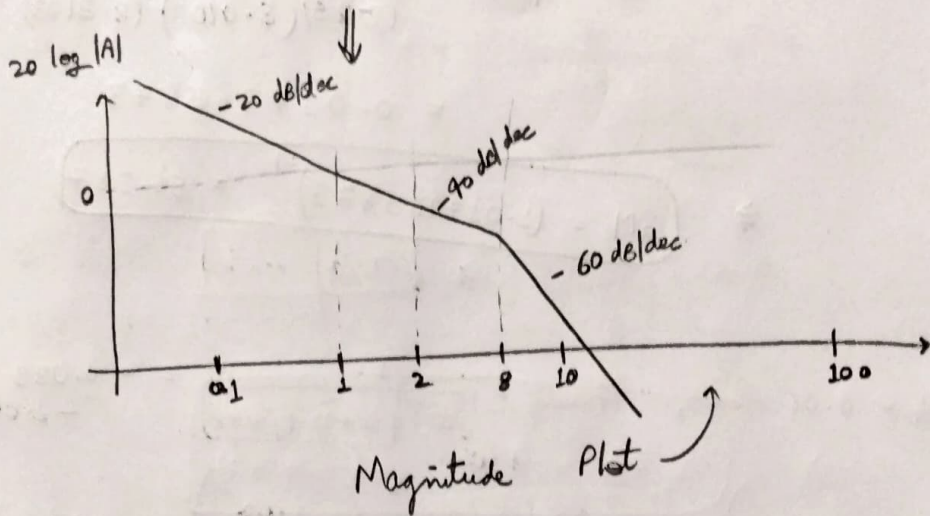
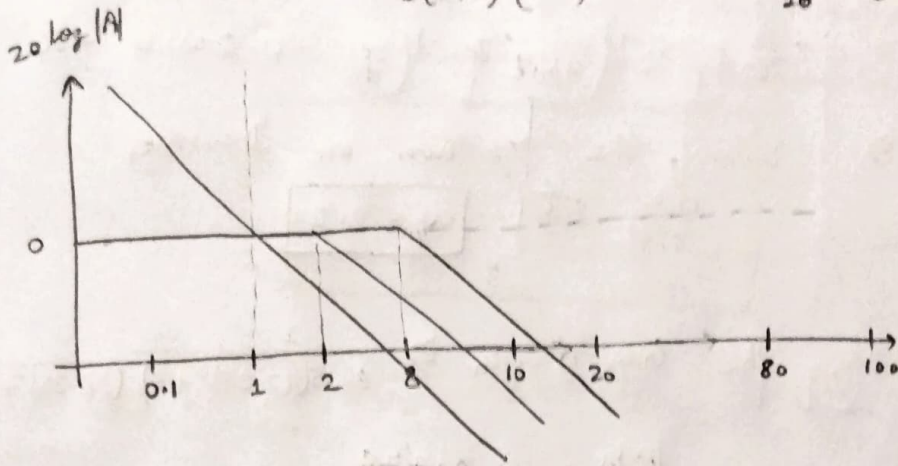
$$\text{at } \omega = \sqrt{2} \quad G'' = -0.5$$

For stable system $-0.5 < \frac{-1}{K} < 0$
 (2 encirclements of $-1/K$)

$\Rightarrow$

$$\boxed{K > 2}$$

⑦ $G(s)H(s) = \frac{K e^{-0.25}}{s(s+2)(s+8)} = \frac{K e^{-0.25}}{16 s \left(\frac{s}{2} + 1\right) \left(\frac{s}{8} + 1\right)}$



due to $e^{-0.25}$

Now, for Gain Crossover $|G(j\omega)| = 1$

$$\Rightarrow \left| \frac{1}{s(s+2)(s+8)} \right| = 1$$

$s = j\omega$ and calculating with calculator, we get

$$\frac{1}{\omega \sqrt{\omega^2 + 4} \sqrt{\omega^2 + 64}} = 1$$

$\Rightarrow$

$$\omega_A = 0.06 \text{ rad/s}$$

= Crossover frequency

Now, for Phase Crossover Frequency,

$$\angle G = -\frac{\pi}{2} - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{8}\right) - 0.2\omega = -\pi$$

$\Rightarrow$ Solving the equation on Desmos,
we get $\boxed{\omega_p = 2.25}$

at $\omega_p = 2.25$, the Gain Margin $\Rightarrow \frac{1}{2.25 \sqrt{(2.25)^2 + 4} \sqrt{(2.25)^2 + 64}}$

$$\Rightarrow |G| = \frac{0.01774}{(2.25)(3.0103)(8.3103)}$$

$$= 0.017765873$$

$$\Rightarrow \boxed{GM = (1.017765873)^{-1} = 56.287}$$

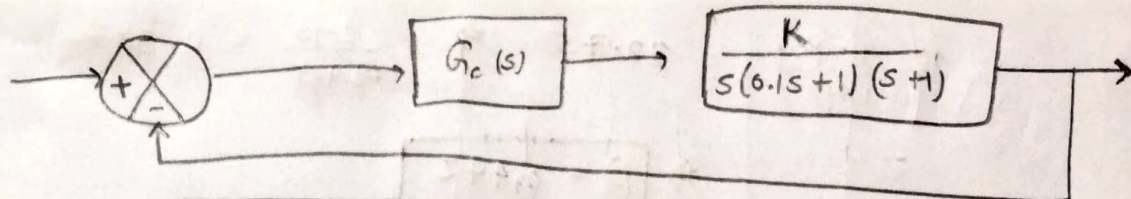
at $\omega_A = 0.06 \text{ rad/s}$, Phase = $\frac{1}{j\omega(j\omega+2)(j\omega+8)} = -0.039 - 1.04j$

$$\Rightarrow \text{Phase Margin} = PM = \tan^{-1}\left(\frac{1.04}{0.039}\right) = 0.2 \times 0.06 \times \frac{180}{\pi}$$

$$= 87.85^\circ - 0.69 = 87.16^\circ$$
$$= PM$$

$$\Rightarrow \boxed{PM = 87.16^\circ}$$

⑧



Let

$$G_c(s) = \frac{1}{\beta} \left(\frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \right)$$

$$K_v = \lim_{s \rightarrow 0} s G_c(s) = \cancel{s} \frac{1}{\beta} \left(\frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \right) \left(\frac{K}{\cancel{s}(0.1s+1)(s+1)} \right)$$

$$= \frac{K}{1} = 4$$

$$\Rightarrow \boxed{K=4}$$

PM = 45° $\Rightarrow \left| j\omega (0.1j\omega+1) (j\omega+1) \right| = 4$

|A| = 1

$$\omega \sqrt{\frac{\omega^2}{100} + 1} \sqrt{\omega^2 + 1} = 4$$

$\omega^2 = x$

$$x^2 \left(\frac{x^2}{100} + 1 \right) (x^2 + 1) = 16$$

$\Rightarrow$ solving for $x \Rightarrow x = 1.5$

$\Rightarrow \omega_A = 1.225$

at $\omega_A = 1.225$

$$\angle G(j\omega) = -90 - \tan^{-1}(1.225) - \tan^{-1}(0.1225)$$

$$= -90 - 50.77 - 6.98$$

$$= -147.75^\circ$$

PM = 180° - 147.75° = 32.25°

$\Rightarrow \phi_{\text{compensator}} = 45 - 32.25 + 10 = 22.75$

↑
for the gain
of the
compensator!

$$\sin^{-1}\left(\frac{1-\beta}{1+\beta}\right) = 22.75 \Rightarrow \frac{1-\beta}{1+\beta} = 0.387$$

$$\Rightarrow \boxed{\beta = 0.442}$$

$$\text{gain added at } \phi_{\max} = \frac{1}{\sqrt{\beta}} = 1.5$$

$$(G|j\omega|) = 1/1.5$$

$$\Rightarrow \frac{4}{\omega \sqrt{1+\omega^2} \sqrt{\frac{\omega^2}{100}+1}} = \frac{1}{1.5} \Rightarrow \omega = 2.317$$

$$\Rightarrow \frac{1}{T\sqrt{\beta}} = 2.317$$

$$\Rightarrow \boxed{T = 0.65}$$

$$\Rightarrow \boxed{G_c(s) = 2.26 \frac{(s+1.538)}{(s+3.477)}}$$

⑨

$$G_c(s) = \frac{\alpha (1 + Ts)}{1 + \alpha Ts}$$

$$s = j\omega$$

$$\Rightarrow \frac{\alpha (1 + j\omega T)}{(1 + \alpha T j\omega)} \times \left(\frac{1 - \alpha T \omega j}{1 - \alpha T \omega j} \right)$$

$$\Rightarrow \left(\frac{\alpha}{1 + \alpha^2 T^2 \omega^2} \right) \left(1 + \omega T (1 - \alpha) j + \alpha \omega^2 T^2 \right)$$

$$= a + bj$$

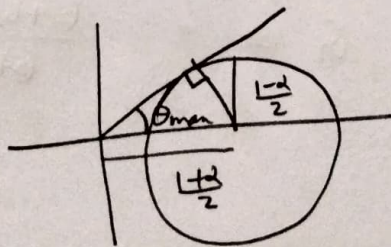
$$a = \frac{\alpha}{1 + \alpha^2 T^2 \omega^2} (1 + \alpha T^2 \omega^2) \quad b = \frac{\omega T (1 - \alpha) \alpha}{1 + \alpha^2 T^2 \omega^2}$$

$$\text{let } \omega T = \lambda \quad a = \frac{\alpha (1 + \alpha \lambda^2)}{1 + \alpha^2 \lambda^2} \quad b = \frac{\alpha \lambda (1 - \alpha)}{1 + \alpha^2 \lambda^2}$$

eliminating λ , we get

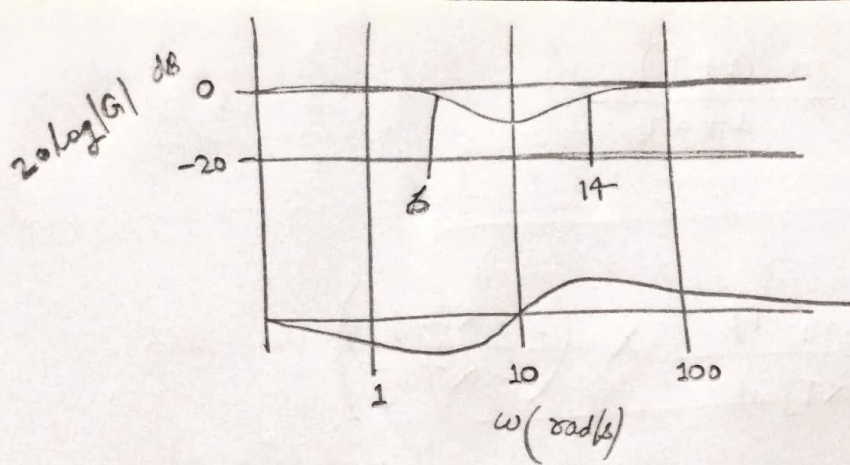
$$\left(a - \frac{1 + \alpha}{2} \right)^2 + b^2 = \left(\frac{1 - \alpha}{2} \right)^2$$

$\Rightarrow$ This is a circle with center $\left(\frac{1 + \alpha}{2}, 0 \right)$ and radius $\frac{(1 - \alpha)^2}{2}$



$$\theta_{\max} = \sin^{-1} \left(\frac{1 - \alpha}{1 + \alpha} \right)$$

④



from graph, we can see $\Rightarrow$ No. of poles = No. of zeros

$$\text{starts from } 0 \text{ dB} \Rightarrow \frac{\prod Z_i}{\prod P_i} = 1$$

From gain plot, it starts falling at $\omega = 1$ (pole) and becomes flat for some $1 < \omega < 10$ (zero) then starts rising for $10 < \omega < 100$ (zero) and finally becomes flat at $\omega = 100$ (pole)

from graph, $Z_1 = 6 \text{ rad/s}$

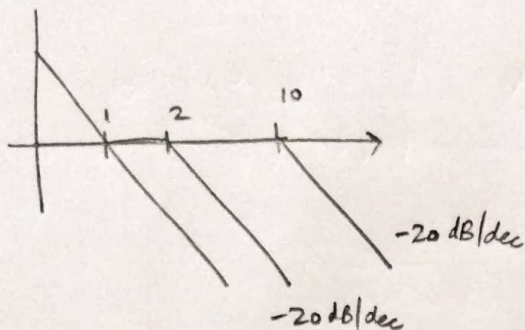
$$\Rightarrow \frac{Z_1 Z_2}{P_1 P_2} = 1 \Rightarrow \frac{6 \times Z_2}{100} = 1$$

$$\Rightarrow Z_2 = 16.\overline{66} \text{ rad/s}$$

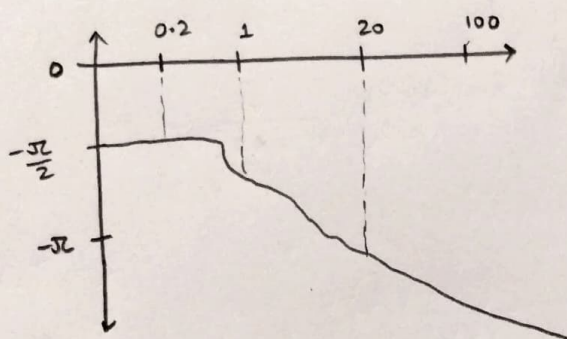
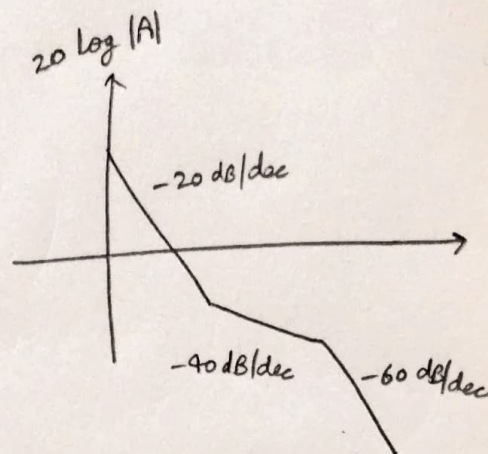
$$\text{TF} \Rightarrow G(s) = \frac{(s+6)(s+16.\overline{66})}{(s+1)(s+100)}$$

⑤

a). $G(s) = \frac{K}{s(0.1s+1)(0.5s+1)}$



$\Rightarrow$



$$\frac{1}{j\omega(1+0.1j\omega)(1+0.5j\omega)} = \frac{1}{j\omega(1+0.6j\omega-0.05\omega^2)}$$

$$1 - 0.05\omega^2 = 0 \Rightarrow \omega = 2\sqrt{5}$$

$$G(2\sqrt{5}) = \frac{-1}{20 \times 0.6} = -\frac{1}{12}$$

For GM 20dB, $G(2\sqrt{5}) = 1/10$

$$\Rightarrow K \times \frac{1}{12} = 1/10 \Rightarrow \boxed{K = 1.2}$$

For $\phi = 45^\circ \Rightarrow 0.6\omega = 1 - 0.05\omega^2$

$$\Rightarrow \omega = 1.4833$$

$$\frac{|G|}{K} = |-0.379 - 0.379j| = 0.536$$

$$\boxed{K = \frac{1}{0.536} = 1.865}$$